Theoretical Spectroscopy via Many-Body Perturbation Theory

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Introduction

The consensus is that the best way to adapt to climate change is to mitigate it by reducing greenhouse gas emissions.[1] To do so requires the development of next-generation green technologies, such as electrocatalysts,[2] photovoltaics,[3] and batteries.[4] These seemingly disparate goals can all be achieved through the discovery of materials with special properties. Spectroscopy is an indispensable tool for experimentalists in the search for such properties, as it illuminates the electronic structure of candidate materials. Computation is crucial for interpreting spectroscopic data, and thus quantum chemists devote much effort to the accurate and efficient simulation of spectra.[5]

Spectroscopy occurs when incoming light perturbs matter, a process which proceeds through excited states of the system. The excited states of a material may be described in terms of quasiparticles, which are collective excitations of interacting electrons. The equation-of-motion coupled-cluster formalism including single and double excitations (EOM-CCSD) can accurately describe these quasiparticles,[?] but it carries a computational scaling of $O(N^6)$, where N is the system size. This is to be compared with semi-local density functional theory's (DFT) $O(N^3)$ scaling, so it comes as no surprise that DFT is the method of choice for large systems.[?] However, DFT is known to be inaccurate for describing excited states,[?] so there is a need for methods that are more accurate than DFT but more computationally tractable than EOM-CCSD.

A compelling alternative is presented by many-body perturbation theory (MBPT).[6] MBPT introduces a variety of "gap-filling" methods, in the sense that these methods provide a more accurate description of excited states than DFT while remaining more computationally tractable than EOM-CCSD.

Excited states via MBPT

The native language of MBPT is that of Green's functions, which are used to describe the propagation of quasiparticles. The exact propagator is given by the interacting Green's function, G. The reference propagator is given by the noninteracting Green's function, G_0. The exact self-energy Σ captures the difference between the two. However, solving for the exact quantities is computationally intractable, so approximations are necessary.

Quasiparticle Energies from GW

GW is a widely used approximation to the self-energy, which gets its name from postulating that Σ is the product of the Green's function G and the screened Coulomb interaction W in the time domain, i.e. $\Sigma^{GW}(t) = iG(t)W(t)$. The GW method has been shown to give quasiparticle energies rivaling EOM-CCSD.[?] Formally, GW carries a computational scaling of $O(N^6)$, but in practice analytic continuation (AC) is used to reduce the scaling to $O(N^4)$. However, GW with AC does a poor job describing quasiparticle energies of core states, which may be important for accurately describing the system.[6]

Via a Löwdin partitioning technique, Loos et al. showed[7] that one can attain the exact quasiparticle energies within the GW approximation by either solving a nonlinear quasiparticle equation, as is traditionally

done, or by determining the eigenvalues of an effective Hamiltonian, which we denote $\mathbf{H}^{\mathrm{GW}}$. However, an explicit construction of $\mathbf{H}^{\mathrm{GW}}$ carries the same $O(N^6)$ cost as solving the exact GW quasiparticle equation.

Moment-conserving GW (MCGW)

Booth et al.[8] claimed to have resolved this issue of diagonalizing $\mathbf{H}^{\mathrm{GW}}$ at a low cost in 2023, by reformulating $\mathbf{H}^{\mathrm{GW}}$ in terms of the spectral moments of the self-energy. Considering self-energy moments to infinite order would be equivalent to constructing the full $\mathbf{H}^{\mathrm{GW}}$, but it is argued that truncating the moment expansion to some finite order yields a good approximation for a portion of the spectrum. A procedure inspired by the Lanczos algorithm is devised to iteratively diagonalize $\mathbf{H}^{\mathrm{GW}}$, with the Ritz values expected to rapidly converge to the ED solution as the number of moments increases. If one uses the resolution-of-the-identity (RI) approximation to the electron repulsion integrals, no step in this procedure scales higher than $O(N^4)$.

However, in my implementation, I found that the Ritz values do not always converge to the exact quasiparticle energies of ED. The Lanczos algorithm is a well-established numerical method[?] whose Ritz values are guaranteed to converge to the exact diagonalization (ED) under proper conditions. However, one of these conditions requires that orthogonality is maintained in the Krylov subspace, which is typically done via periodic reorthogonalization.

Figure 1 shows, for a number of small molecules, that MCGW initially follows the expected Lanczos convergence behavior towards ED, but then diverges. Analysis reveals that MCGW does not have access to its own Krylov subspace and therefore cannot perform reorthogonalization, leading to numerical instability. Therefore, it is misleading to associate MCGW with Lanczos. A reliable quantum chemistry method should remain faithful to its theoretical foundation without ambiguity from numerical artifacts, ensuring that errors remain below other dominant sources such as basis set incompleteness and grid effects.[?]

Spectral function of the uniform electron gas (UEG)

The UEG is the paradigmatic metallic system,[?] serving as a testbed for many-body methods hoping to describe metals. Its exact spectral function in the thermodynamic limit (TDL) is not known. Green's function methods are well suited for computing the spectral function $A(\omega)$, since the spectral function is proportional to the imaginary part of the retarded Green's function:

$$A(\omega) = -\frac{1}{\pi}\mathrm{Im}, G^R(\omega). \tag{1}$$

Now, the retarded Green's function satisfies the Dyson equation:

$$G^R(\omega) = G_0^R(\omega) + G_0^R(\omega)\Sigma^R(\omega)G^R(\omega). \tag{2}$$

Because obtaining the exact self-energy is intractable, MBPT methods approximate Σ through a perturbative expansion, with the interaction strength as the perturbation parameter. The GW approximation corresponds to choosing the parameter as the screened Coulomb interaction W and truncating at first order in the perturbation series. This is equivalent to inserting Σ^{GW} into this equation. However, solving the Dyson equation through truncation at first order fails to capture satellite features in core-level spectral functions of the UEG.[? ? ?]

In contrast, the cumulant expansion is able to reproduce satellite features. It expresses the diagonal retarded Green's function for a quasiparticle p as

$$G_{pp}^R(t) = G_{0,pp}^R(t)e^{C_{pp}^R(t)}, \tag{3}$$

where the retarded cumulant, with quasi-boson frequencies ω, is given by

$$C_{pp}^R(t) = -\frac{1}{\pi}\int d\omega \frac{\mathrm{Im}\Sigma_{pp}^R(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right). \tag{4}$$

This form can be derived by solving the large polaron problem, where a core hole is coupled to a bath of bosons.[?]

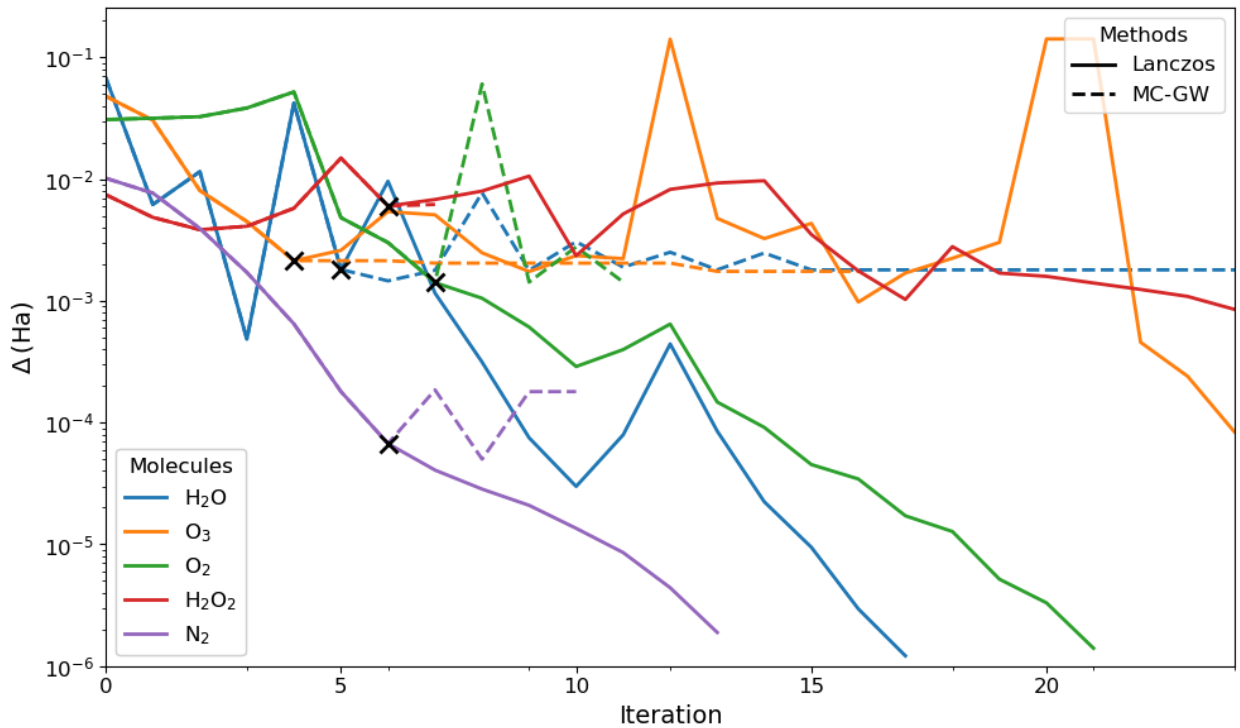


Figure 1: Convergence behaviour of MCGW for the quasiparticle energies of a variety of small molecules in the cc-pVDZ basis. Δ is the maximum difference between the ED solution and the Ritz value at a given Lanczos iteration in the pseudo-active space. This pseudo-active space is defined by the combination of all N_{occ} orbitals plus the first N_{occ} virtual orbitals. The location of the desired numerical accuracy of a quantum chemistry method is at the bottom of the plot, or 1×10^{-6} Hartrees. MCGW iteration was terminated one the maximum overlap between the Krylov vectors surpassed 1×10^{-4}, as at this point is it known that the Ritz values will not converge any further towards the ED solution.

GW+C

Initial cumulant approaches used Σ^{GW} to construct the cumulant.[?] This has been attractive because the GW self-energy may be built at a cost of $O(N^4)$, and the cumulant can be constructed from Σ^{GW} without incurring any additional cost. While this approach produces satellite features, their positions occur at incorrect locations relative to EOM-CCSD references.[?]

PT2+C

Second-order perturbation theory (PT2) provides a second-order self-energy, whose construction costs $O(N^5)$, which can be used to build a cumulant as

$$C^{(2)}_{pp}(t) = \frac{1}{2}\sum_{iab} \frac{\langle pi \| ab \rangle^2}{\left(\epsilon^{ab}_{pi}\right)^2} \left(e^{-i\epsilon^{ab}_{pi} t} + i\epsilon^{ab}_{pi} t - 1\right) + \frac{1}{2}\sum_{ija} \frac{\langle pa \| ij \rangle^2}{\left(\epsilon^{ij}_{pa}\right)^2} \left(e^{-i\epsilon^{ij}_{pa} t} + i\epsilon^{ij}_{pa} t - 1\right). \tag{5}$$

The quasi-boson frequencies are defined as $\epsilon^{ab}_{pi} = \varepsilon_p - \varepsilon_i + \varepsilon_a - \varepsilon_b$ and $\epsilon^{ij}_{pa} = \varepsilon_p + \varepsilon_a - \varepsilon_i - \varepsilon_j$, with ε being the Hartree-Fock orbital energies. This would seem to offer an alternative to GW+C, but in metallic systems like the UEG, PT2 diverges in the thermodynamic limit,[?] rendering the approach void.

BWs2+C

These issues may be mitigated by regularizing the second-order self-energy. With this purpose in mind, Head-Gordon developed[? ?] a size-consistent Brillouin–Wigner second-order perturbation theory (BWs2)

method, which scales as $O(N^5)$. In PT2, one splits the Hamiltonian into a zeroth order part, $\hat{H}^{(0)}$, and a perturbation, $\hat{V}$, as

$$\hat{H} = \hat{H}^{(0)} + \hat{V}. \tag{6}$$

Denoting general indices as $p, q, \ldots$, occupied indices as $i, j, \ldots$, and virtual indices as $a, b, \ldots$, the Moller-Plesset partitioning takes

$$\hat{H}^{(0)} = f_{pq}\hat{a}_p^\dagger \hat{a}_q, \tag{7}$$

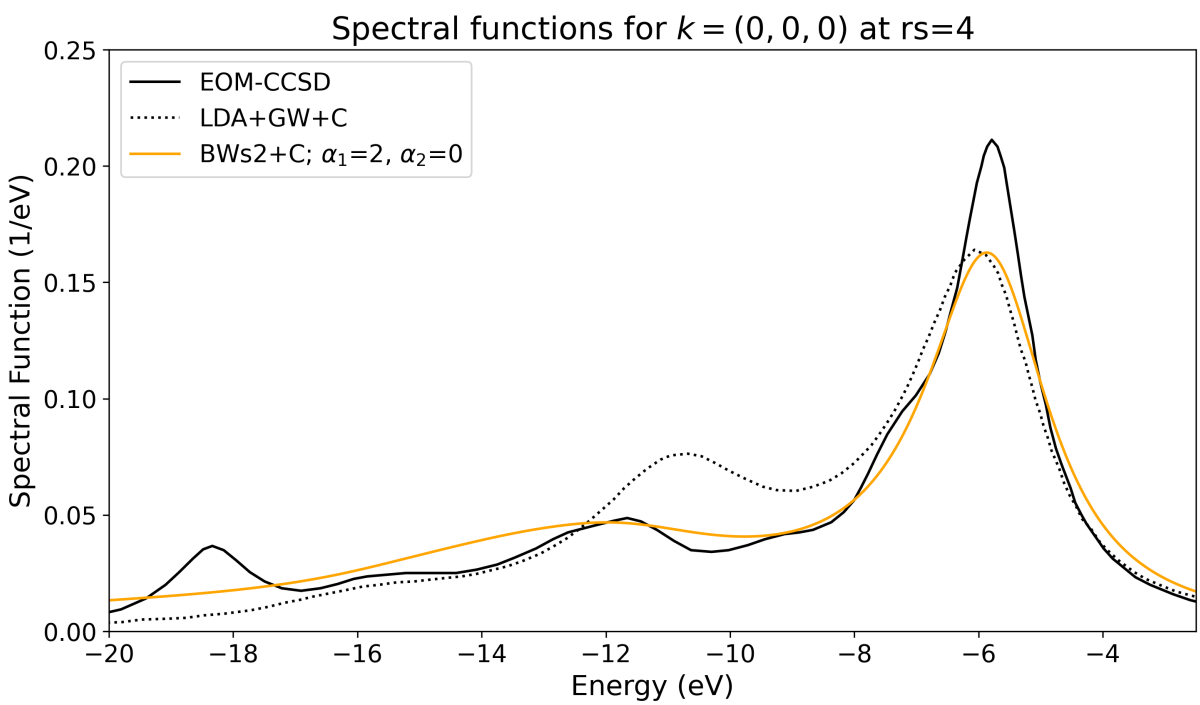
where f_{pq} is the Fock matrix. In contrast, the size-consistent Brillouin-Wigner paritioning takes

$$\hat{H}^{(0)} = f_{pq}\hat{a}_p^\dagger \hat{a}_q + \frac{\alpha_1}{8}\left(t_{ik}^{ab(1)}\langle jk\|ab\rangle + t_{jk}^{ab(1)}\langle ik\|ab\rangle - \frac{2}{n_e}t_{kl}^{ab(1)}\langle kl\|ab\rangle\delta_{ij}\right)\hat{a}_i^\dagger \hat{a}_j + \frac{\alpha_2}{8}\left(t_{ij}^{ac(1)}\langle ij\|bc\rangle + t_{ij}^{bc(1)}\langle ij\|ac\rangle\right)\hat{a}_a^\dagger \hat{a}_b, \tag{8}$$

where $t^{(1)}$ are the doubles amplitudes of the perturbation theory to first order, n_e is the number of electrons, and α_1 and α_2 are empirical parameters. BWs2 has been shown to accurately compute ground state properties of metals.[?] Recently, Tew[?] put the BWs2 self-energy on firm theoretical grounds, by showing that the theory admits a Goldstone self energy, instead of the Dyson one used by methods, including GW. This implies that BWs2 obeys a Dyson equation like equation 2. However, G_0 is no longer due to HF, but rather to the "dressed" BWs2 input. This led me to derive the following form for the BWs2 cumulant as

$$C_{pp}^{\mathrm{BWs2}}(t) = \frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2}\left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} - 1\right), \tag{9}$$

where $\tilde{\epsilon}_{pi}^{ab} = \tilde{\varepsilon}_p - \tilde{\varepsilon}_i + \tilde{\varepsilon}_a - \tilde{\varepsilon}_b$ and $\tilde{\epsilon}_{pa}^{ij} = \tilde{\varepsilon}_p + \tilde{\varepsilon}_a - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j$, with $\tilde{\varepsilon}$ being the single-particle energies due to the dressed BWs2 input. Preliminary results, shown in figure , suggest that the fifth order scaling BWs2+C may improve upon the fourth order scaling GW+C, without incurring the $O(N^6)$ cost of EOM-CCSD reference. The study of larger UEG supercells is underway, which may demonstrate that BWs2+C is able to provide a viable alternative to GW+C, when EOM-CCSD is not computationally feasible.



Future directions

A self-consistent cumulant expansion

Self-consistency has been incorporated into Dyson-based approaches with mixed success. These approaches solve the Dyson equation perturbatively, so adding self consistency may lead to the double counting of terms in the expansion. Pursuing self-consistency within the cumulant framework may be theoretically advantageous, where diagrammatic consistency may be less restrictive. There have been a few studies of the self-consistent cumulant, but they have all been done analytically in the TDL.[?] [?] [?] We propose to pursue a numerically self-consistent cumulant. This can be understood by defining a frequency grid to solve for $C^R(t)$ via equation 4, which may then be used to obtain $G^R(\omega)$ via equation 3. Then, the Dyson equation can be rearranged to solve for $\Sigma^R(\omega)$ as

$$\Sigma^R(\omega) = G_0^R(\omega)^{-1} - G^R(\omega)^{-1}, \tag{10}$$

which yields an updated $\Sigma^R(\omega)$ that may be used to update $C^R(t)$, and so on until convergence.

Memory kernel as an alternative to the self-energy

The generalized quantum master equation (GQME) describes the time evolution of a correlation function $\mathcal{C}(t)$ as

$$\dot{\mathcal{C}}(t) = \mathcal{C}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{C}(t-\tau)\mathcal{K}_1(\tau) + D(t), \tag{11}$$

with the moment Ω_1, the memory kernel $\mathcal{K}_1(t)$, and an inhomogeneous term $D(t)$, which may be set to zero for equilibrium correlation functions.[?] This is to be compared with the equation of motion of the retarded Green's function, which is given by

$$i\partial_t G^R(t) = \epsilon_0 G^R(t) + \int_0^t d\tau\, \Sigma^R(t-\tau)\, G^R(\tau), \tag{12}$$

where ϵ_0 is the noninteracting quasiparticle energy. This implies that $\mathcal{K}_1(t)$ and $\Sigma^R(t)$ may both be used to obtain a perturbative solution to equation 2. The difference is in the choice of perturbative parameter; in the former, a perturbative expansion is carried out in the time variable, while in the latter one expands in the interaction strength. The memory kernel is expected to decay quickly with time,[3?] while the self-energy may not decay quickly with the interaction strength.[?] A symptom of the latter phenomenon is an inability of perturbative expansions in the interaction strength to adequately describe strongly correlated systems, where the interaction is large. It happens to be the case that strong correlated materials have been shown to exhibit special properties. Thus, the development of memory kernel based methods has the potential to accelerated materials discovery for climate change mitigation.

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